
بسم الله الرحمن الرحيم

اَللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَّافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُّتَقَبَّلًا،

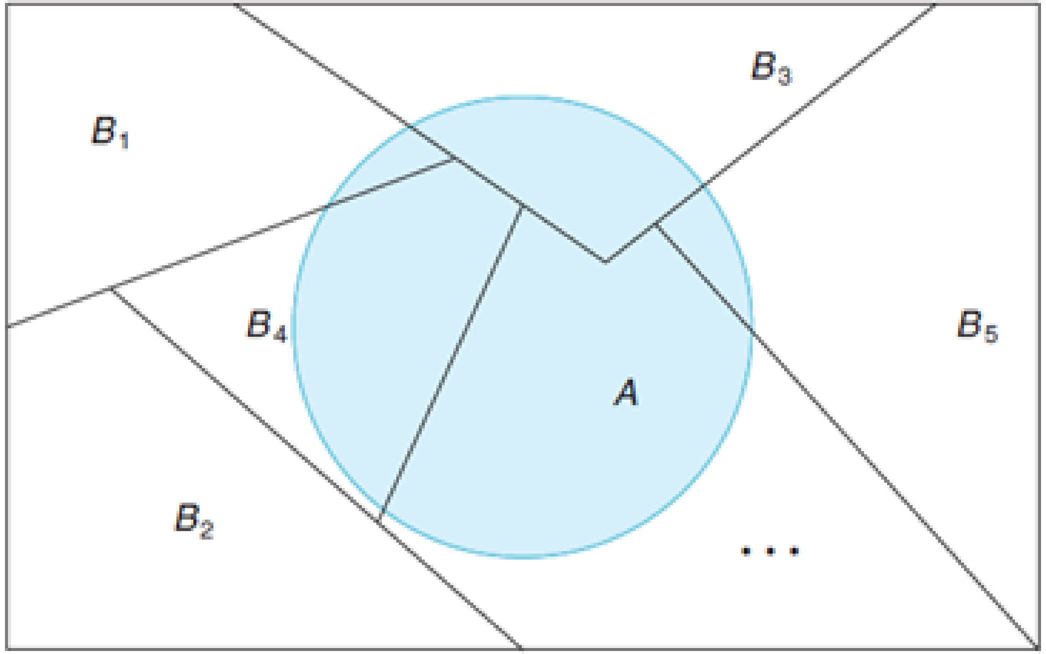
Theorem on Total Probability and Bayes Law

Week 5

Theorem on Total Probability/ Rule of Elimination

If the events $B_1, B_2, \ldots, B_k$ constitute a partition of the sample space S such that $P(B_i) \neq 0$ for $i = 1, 2, \ldots, k$, then for any event A of S,

$$P(A) = \sum_{i=1}^{k} P(B_i \cap A) = \sum_{i=1}^{k} P(B_i)P(A|B_i).$$

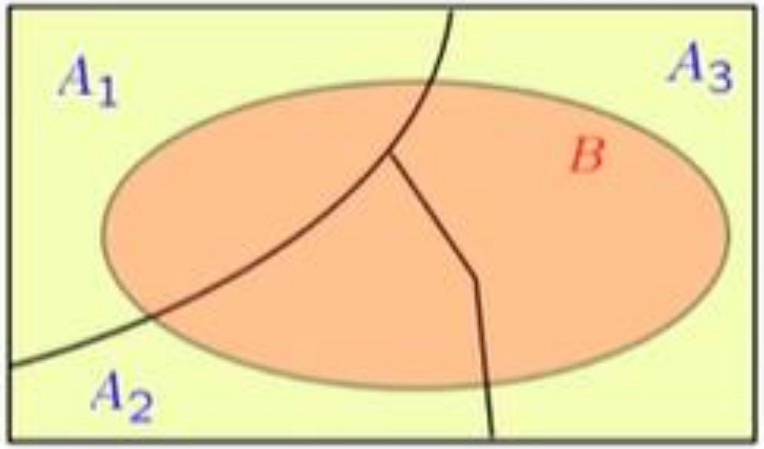


Important point:
Bi`s must be

i. **mutually exclusive**
ii. **Collectively exhaustive**.

Total Probability

$$\boxed{(B) = (A_1 \cap B) + (A_2 \cap B) + (A_3 \cap B)}$$

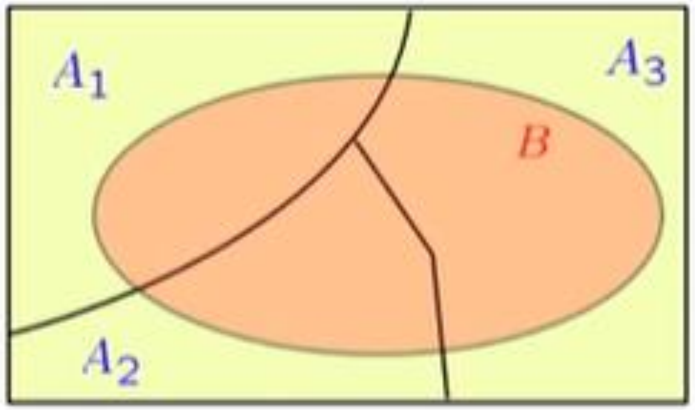


$$\boxed{P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B)}$$

$$\boxed{P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}$$

Example: 1

In a certain assembly plant, three machines, *A*1, *A*2, and *A*3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability that it is defective?



Sol: Let B denote that product is defective.

P(A1)=0.30, P(A2)=0.45,P(A3)=0.25

P(B|A1)=0.02, P(B|A2)=0.03, P(B|A3)=0.02

P(B)=?

P(B)=P(A1) P(B|A1)+ P(A2) P(B|A2)+ P(A3) P(B|A3)

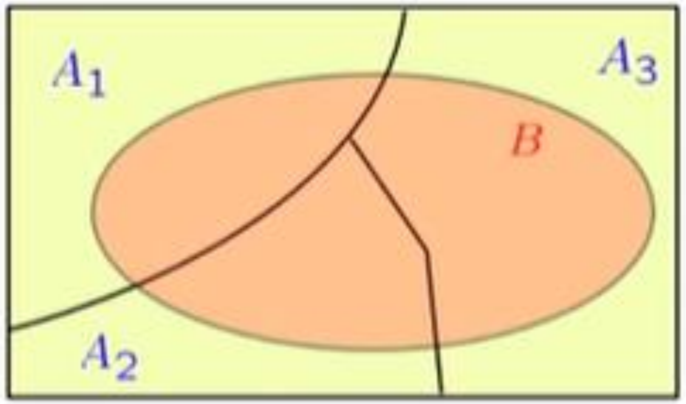
P(B)= (0.30)(0.02)+(0.45)(0.03)+(0.25)(0.02)

Example: 2

The flooding of a river in the spring season will depend on the accumulation of snow in the mountains during the past winter season. The accumulation of snow may be described as Heavy (H), Normal (N) and Light (L). Assume here that flooding depends on the accumulation of snow. Also P(F|H)=0.9, P(F|N)=0.4, P(F|L)=0.1, Where as in a given winter season P(H)=0.2, P(N)=0.5, P(L)=0.3. Find the prob. of flooding in the river during the following spring season.

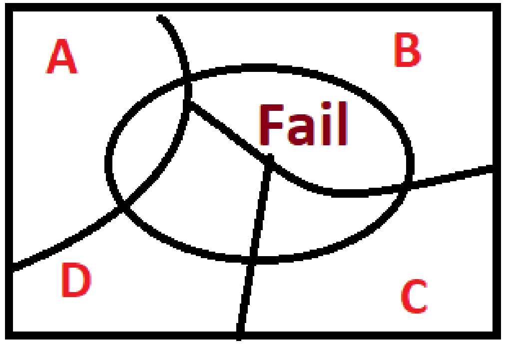
$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$\begin{aligned} P(\text{Flood}) &= P(H)P(F|H) + P(N)P(F|N) + P(L)P(F|L) \\ &= (0.20)(0.90) + (0.50)(0.40) + (0.30)(0.10) \end{aligned}$$



Example: 3

After production, an electrical circuit is given a quality score of A, B, C, or D. Over a certain period of time, 77% of the circuits were given a quality score A, 11% were given a quality score B, 7% were given a quality score C, and 5% were given a quality score D. Furthermore, it was found that 2% of the circuits given a quality score A eventually failed, and the failure rate was 10% for circuits given a quality score B, 14% for circuits given a quality score C, and 25% for circuits given a quality score D. Find the probability of failure of circuit.



$$P(\text{fail}) = P(A \cap \text{fail}) + P(B \cap \text{fail}) + P(C \cap \text{fail}) + P(D \cap \text{fail})$$

$$= P(A)P(\text{fail} \mid A) + P(B)P(\text{fail} \mid B) + P(C)P(\text{fail} \mid C) + P(D)P(\text{fail} \mid D)$$

$$= (0.77)(0.02) + (0.11)(0.10) + (0.07)(0.14) + (0.05)(0.25)$$

$$= 0.0487$$

Home Task-1

In a bolt factory, machines A, B and C manufacture 25, 60 and 15 percent of the total output respectively. Their outputs 5, 4 and 2 percent respectively are defective bolts.

Find the probability that a bolt selected at random is defective.

Home Task 2

Bowl 1 contains 4 red chips, Bowl 2 contains 3 white chips and Bowl 3 contains 2 red and 5 white chip. A bowl is drawn at random and one chip is drawn at random from that bowl.

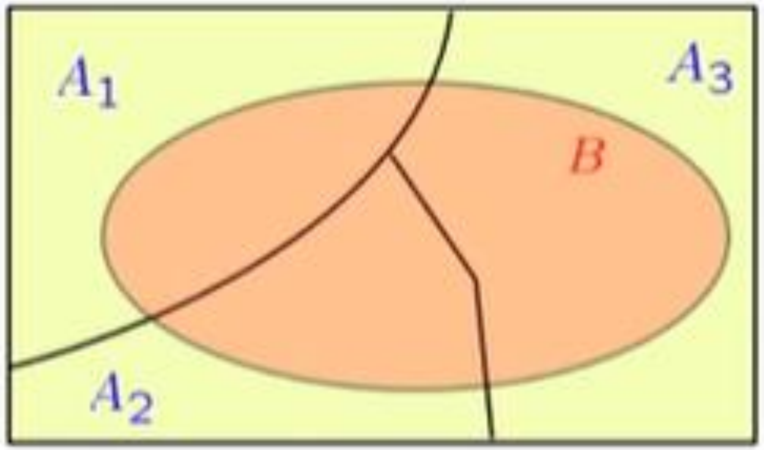
Compute the probability of selecting a white chip.

Home Task 3

Bowl 1 contains 4 red chips, Bowl 2 contains 3 white chips, Bowl 3 contains 6 red chips and 2 white chips and Bowl 4 contains 2 red and 5 white chips. A bowl is drawn at random and one chip is drawn at random from that bowl.

Compute the probability of selecting a red chip.

Total Probability & Bayes' Law



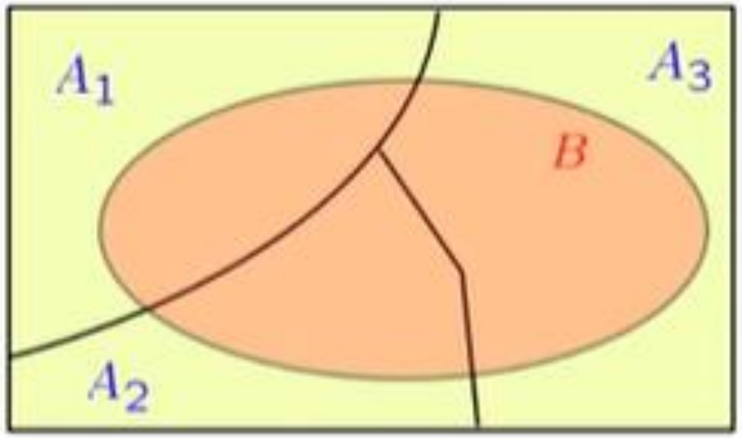
$$\boxed{P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)}$$

$$\boxed{P(A_i \mid B) = ?}$$

Bayes' Law

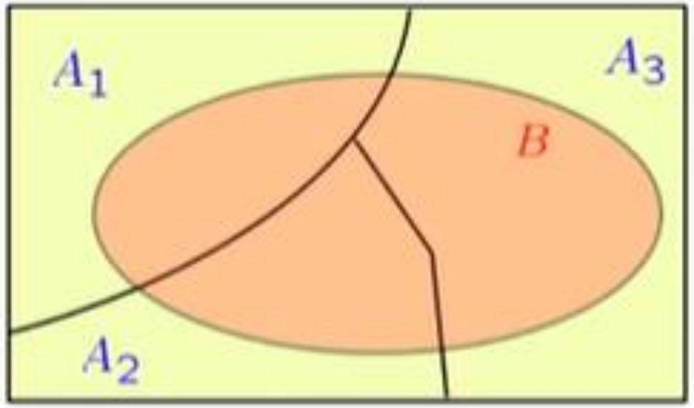
If the events A1, A2,… Ak constitute a partition of the sample space S such that $P(Ai) \neq 0$ for any i=1,2,…,k, then for any event B in S such that $P(B) \neq 0$,

$$P(Ai|B) = \frac{P(Ai)P(B|Ai)}{P(B)}$$



Example: 1

In a certain assembly plant, three machines, *A*1, *A*2, and *A*3, make 30%, 45%, and 25%, respectively, of the products. It is known from past experience that 2%, 3%, and 2% of the products made by each machine, respectively, are defective. Now, suppose that a finished product is randomly selected. What is the probability defective bolt came from machine A2?



Sol: Let B denote that product is defective.

$$P(A2|B) = ?$$

$$P(A1) = 0.30, P(A2) = 0.45, P(A3) = 0.25,$$

$$P(B|A1) = 0.02, P(B|A2) = 0.03, P(B|A3) = 0.02$$

$$P(B) = P(A1)P(B|A1) + P(A2)P(B|A2) + P(A3)P(B|A3)$$

$$= (0.30)(0.02) + (0.45)(0.03) + (0.25)(0.02)$$

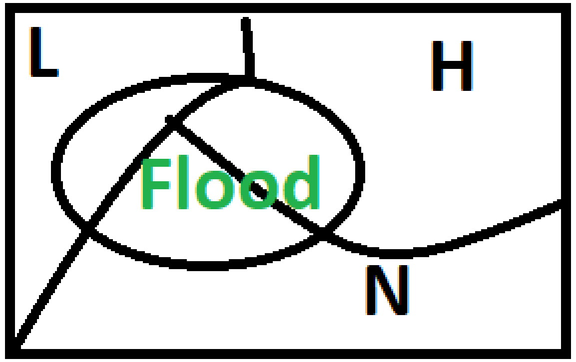
$$P(A2|B) = \frac{P(A2)P(B|A2)}{P(B)}$$

$$= \frac{(0.45)(0.03)}{(0.30)(0.02) + (0.45)(0.03) + (0.25)(0.02)}$$

Example: 2

The flooding of a river in the spring season will depend on the accumulation of snow in the mountains during the past winter season. The accumulation of snow may be described as Heavy (H), Normal (N) and Light (L). Assume here that flooding depends on the accumulation of snow. Also P(F|H)=0.9, P(F|N)=0.4, P(F|L)=0.1, Where as in a given winter season P(H)=0.2, P(N)=0.5, P(L)=0.3. Find the prob. of heavy accumulation of snow given that river is facing flood.

$$P(\text{Flood}) = P(H)P(F|H) + P(N)P(F|N) + P(L)P(F|L)$$
$$= (0.20)(0.90) + (0.50)(0.40) + (0.30)(0.10)$$



$$P(H \mid F) = \frac{P(H)P(F \mid H)}{P(F)}$$
$$= \frac{(0.20)(0.90)}{(0.20)(0.90) + (0.50)(0.40) + (0.30)(0.10)}$$

$$P(A_i \mid B) = \frac{P(A_i)P(B \mid A_i)}{P(B)}$$

$$P(B) = P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)$$

$$P(H \mid F) = ?$$

Example: 3

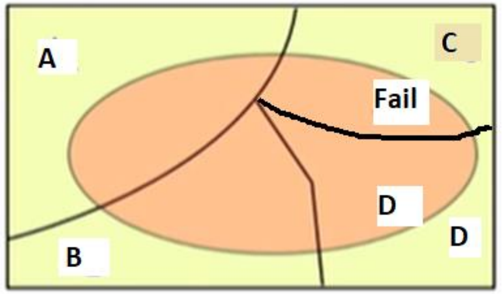
After production, an electrical circuit is given a quality score of A, B, C, or D. Over a certain period of time, 77% of the circuits were given a quality score A, 11% were given a quality score B, 7% were given a quality score C, and 5% were given a quality score D. Furthermore, it was found that 2% of the circuits given a quality score A eventually failed, and the failure rate was 10% for circuits given a quality score B, 14% for circuits given a quality score C, and 25% for circuits given a quality score D. If a circuit failed, what is the probability that it had received a quality score either C or D?

$$
\begin{aligned}
P(\text{fail}) &= P(A \cap \text{fail}) + P(B \cap \text{fail}) + P(C \cap \text{fail}) + P(D \cap \text{fail}) \\
&= P(A)P(\text{fail} \mid A) + P(B)P(\text{fail} \mid B) + P(C)P(\text{fail} \mid C) + P(D)P(\text{fail} \mid D) \\
&= (0.77)(0.02) + (0.11)(0.10) + (0.07)(0.14) + (0.05)(0.25) \\
&= 0.0487
\end{aligned}
$$

$$
\begin{aligned}
P(C \mid \text{fail}) &= \frac{P(C)P(\text{fail} \mid C)}{P(\text{fail})} \\
&= 0.2012
\end{aligned}
$$

$$
\begin{aligned}
P(D \mid \text{fail}) &= \frac{P(D)P(\text{fail} \mid D)}{P(\text{fail})} \\
&= 0.2567
\end{aligned}
$$

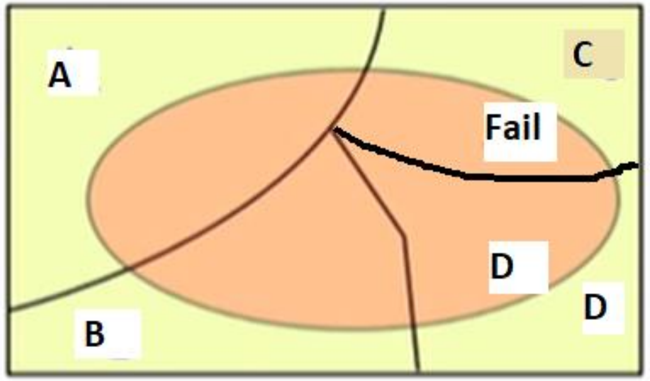
$$
\boxed{P(C \mid \text{fail}) + P(D \mid \text{fail}) = 0.4579}
$$



Example: 3

After production, an electrical circuit is given a quality score of A, B, C, or D. Over a certain period of time, 77% of the circuits were given a quality score A, 11% were given a quality score B, 7% were given a quality score C, and 5% were given a quality score D. Furthermore, it was found that 2% of the circuits given a quality score A eventually failed, and the failure rate was 10% for circuits given a quality score B, 14% for circuits given a quality score C, and 25% for circuits given a quality score D. If a circuit did not fail, what is the probability that it had received a quality score A?

$$
\begin{aligned}
P(A \mid \text{not fail}) &= \frac{P(A)P(\text{not fail} \mid A)}{P(\text{not fail})} \\
&= \frac{(0.77)(1-0.02)}{(1-0.0487)} \\
&= 0.7932
\end{aligned}
$$



$$
\begin{aligned}
P(\text{fail}) &= P(A \cap \text{fail}) + P(B \cap \text{fail}) + P(C \cap \text{fail}) + P(D \cap \text{fail}) \\
&= P(A)P(\text{fail} \mid A) + P(B)P(\text{fail} \mid B) + P(C)P(\text{fail} \mid C) + P(D)P(\text{fail} \mid D) \\
&= (0.77)(0.02) + (0.11)(0.10) + (0.07)(0.14) + (0.05)(0.25) \\
&= 0.0487
\end{aligned}
$$

Home Task-1

In a bolt factory, machines A, B and C manufacture 25, 60 and 15 percent of the total output respectively. Their outputs 5, 4 and 2 percent respectively are defective bolts. A bolt is selected at random and found to be defective. What is the probability that the bolt came from

i. Machine A?
ii. Machine B?
iii. Machine C?

Home Task 2

Bowl 1 contains 4 red chips, Bowl 2 contains 3 white chips and Bowl 3 contains 2 red and 5 white chip. A bowl is drawn at random and one chip is drawn at random from that bowl.

i. Compute the probability of selecting a white chip.

ii. If the selected chip is white, compute the conditional probability that the other chip in the bowl is red (Bowl 3 is selected).

Home Task 3

Aggregates for the construction of a reinforced concrete building are supplied by two companies (A, B). Orders for company A to deliver 600 truck loads a day and 400 truck loads a day for company B. From prior experience, it is expected that 3% and 1% of the material will be substandard for company A and B respectively. If a load of aggregates is found to be substandard, what is the probability that it is from company A?

عشق قاتل سے بھی، مقتول سے ہمدردی بھی
یہ بتا کس سے محبت کی جزا مانگے گا ؟

سجدہ خالق کو بھی، ابلیس سے یارانہ بھی
حشر میں کس سے عقیدت کا صلہ مانگے گا ؟

ڈاکٹر علامہ محمد اقبال